



Bias & Variance in LLM-as-a-Judge

Single judge, repeated judge, or panel? What the recent literature says about which strategy reduces which error — and why it matters for evaluation budgets.

Research · 2026-05-22 8 slides ~6 min

THE PROBLEM

A single judge carries systematic biases

- **Position bias**

Order of candidates flips outcomes. Not random — varies by judge and task; modulated by quality gap (Shi et al. 2025).

- **Verbosity bias**

Longer answers preferred regardless of substance.

- **Self-preference**

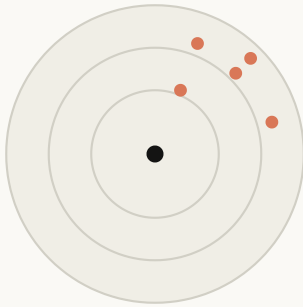
Judges favor outputs from their own model family (Zheng 2023, Verga 2024).

- **Reasoning ceiling**

Judge inherits its own blind spots — math, logic, edge cases.

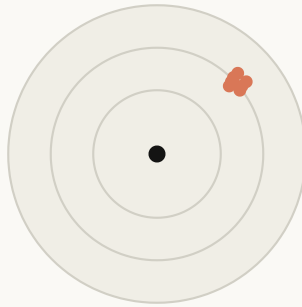
MENTAL MODEL

Four targets, four strategies



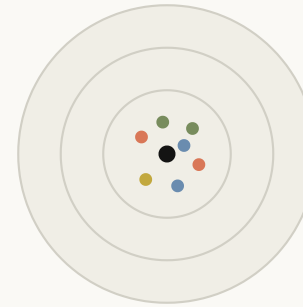
Single judge, 1 call

high bias
high variance



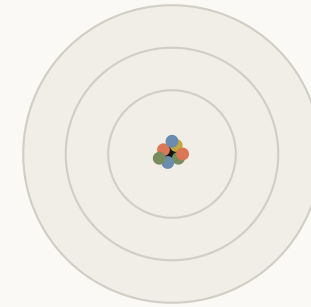
Single, repeated

high bias
low variance



Panel (PoLL)

low bias
medium variance



CyclicJudge

low bias
low variance

Repetition tightens the cluster. Diverse panel pulls it toward the bullseye. Balanced rotation does both — for the same budget.

THE KEY INSIGHT

Repetition $\neq$ bias correction

Adding more prompts or generations with the *same* judge shrinks sampling noise — not judge bias.

Zhu et al. (2026) show the judge-identity variance component is independent of sample count. To shrink it, you must change K — the number of distinct judges. This is the bias-variance tradeoff, stated cleanly for LLM evaluators.

WHY — THE DECOMPOSITION

$$\begin{aligned} X_{ij\ell} = & \mu_{\theta} \\ & + \alpha_i \quad \text{scenario} \\ & + \beta_{ij} \quad \text{generation} \end{aligned}$$

+ γ_ℓ judge bias

+ $\varepsilon_{ij\ell}$ residual

- shrinks with more scenarios
- shrinks with more generations
- only** shrinks with more judges
- shrinks with everything

Var(score)

THREE PANEL STRATEGIES

Aggregation matters as much as diversity

- **PoLL — small diverse jury, majority vote**

Verga et al. 2024. Three judges from disjoint families beat single GPT-4 on 6 datasets at ~7× lower cost. Bias cancellation depends on family diversity.

- **BT- σ — reliability-weighted aggregation**

Qian et al. 2026. Bradley-Terry extended with per-judge discriminator, learned unsupervised. Beats naive averaging when judges differ in quality.

- **CyclicJudge — balanced round-robin**

Zhu et al. 2026. Each judge scores each cell at most once, equal usage overall. Cancels bias in expectation AND minimizes variance at fixed budget.

γ stays

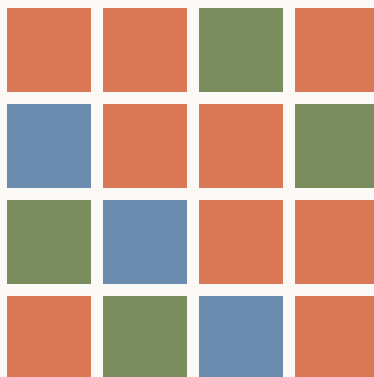
1 judge
1 call

1 judge
repeated

panel
round-robin

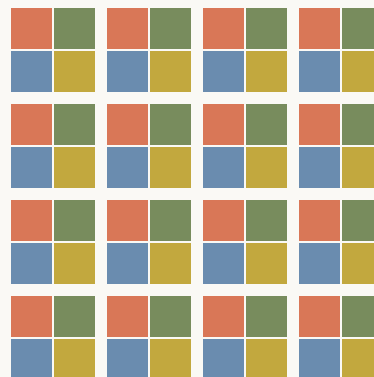
DESIGN COMPARISON

Three ways to spend the same budget



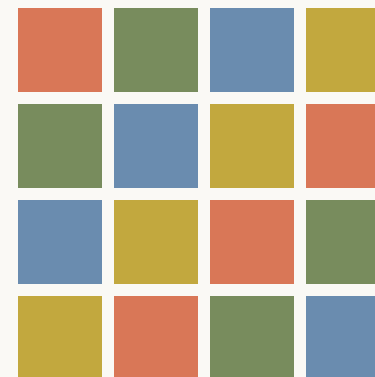
Random single judge

one judge per cell,
chosen at random
→ unbalanced, biased



All judges on all cells

unbiased,
but 4× the cost
→ fewer scenarios fit budget



CyclicJudge (round-robin)

each judge used equally,
same cost as random
→ **27–40% lower variance**

Rows = scenarios · Columns = generations · Cell color = judge used

FOR OUR EVAL FRAMEWORK

How to spend the budget

- Internal iteration → single strong judge + position swap
Cheap. Sufficient for relative deltas under a held-fixed judge.
- Leaderboards & cross-provider comparison → CyclicJudge round-robin
≥3 judges from disjoint families. Same call budget, ~30% lower variance, unbiased ranking.
- Heterogeneous judge quality → BT- σ weighted aggregation
When mixing frontier + small open-weight judges. Learns reliability unsupervised.
- Always report the judge(s)
Judge identity is now understood to be the dominant variance source. Pin versions.

Refs: Zhu 2026 (CyclicJudge) · Verga 2024 (PoLL) · Qian 2026 (BT- σ) · Shi 2025 (position bias) · Zheng 2023 (MT-Bench)